Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

● **Data Loading**: Imported the dataset using pandas.
● **Initial Exploration**: Used df.info() to check structure and .describe() for summary statistics.

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

● **Missing Data Handling**: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
● **Column Standardization**: Renamed columns to snake case for better readability and documentation.
● **Feature Engineering:** ○ Created age_group column by binning customer ages. ○ Created purchase_frequency_days column from purchase data.
● **Data Consistency Check**: Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
● **Database Integration**: Connected Python script to MYSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MYSQL to answer key business questions:

1. Revenue by Gender – Compared total revenue generated by male vs. female customers.

gender	revenue
Male	157890
Female	75191

2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88

3. Top 5 Products by Rating – Found products with the highest average review ratings.

product	avg_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

shipping_type	avg_purchase_amount
Express	60.48
Standard	58.46

5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
No	2847	59.87	170436
Yes	1053	59.49	62645

6. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

product	with_discount_purchase_percentage
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37
Boots	46.53
Jeans	45.97
Dress	45.18
Hoodie	45.03
Backpack	44.76
Belt	44.72
Jewelry	44.44
Shorts	43.31

7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	Number of customers
Loyal	3116
Returning	701
New	83

8. Top 3 Products per Category – Listed the most purchased products within each category.

product_rank	category	product	purchase_count
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
Yes	958
No	2518

10. Revenue by Age Group – Calculated total revenue contribution of each age group.

age_group	revenue
young_adult	62143
middle-aged	59197
adult	55978
senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.

6. Insights

- While analysing revenue by gender we found that revenue by females are less compared to males by almost half, there is an untapped growth opportunity in the female segment.

- Total revenue and average spend by the customers were independent of subscription as the non-subscribed customers spent more on average and had contributed more to the total revenue than the customers who had subscription.

- There are many returning/repeating buyers where we have the opportunity to move them into our Loyal segment.

- We have many products which are discount dependent that are being sold almost 40-50% times when discount is applied, so discount are driving sales volume

- We have found top products performing in each category with high sales in their category, hence they can be used as a part of marketing strategy

7. Business Recommendations

● **Boost Subscriptions** – Promote exclusive benefits for subscribers.

● **Opportunity to Grow Revenue from Female Customers** - Business should introduce targeted marketing campaigns, personalized promotions, and product strategies tailored toward female preferences to increase engagement, sales and revenue.

● **Customer Loyalty Programs** – Reward repeating or returning buyers to move them into the “Loyal” segment.

● **Review Discount Policy** – Balance sales boosts with margin control.

● **Product Positioning** – Highlight top-rated and best-selling products in campaigns.

● **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.